RestoNiti

Cloud services brief - architect review

Proposed | 2026-09-12 | Keep short for review

Decision in one line

Azure App Service (Linux) + Azure Database for PostgreSQL Flexible Server + Cloudflare R2 (+ Cloudflare Pages later for QR / owner web). Not Azure SQL for production. POS stays offline SQLite.

1. Service map

Service	Role	Used for
App Service Linux B1 (-> B2)	.NET 10 Cloud API	Activate, recover, backup upload/list; later owner APIs
Postgres Flexible B1ms (-> B2s), 32 GB, Central India	Shared multi-tenant DB	Backup metadata; later owner read models / events
Cloudflare R2 (private)	Object storage	Encrypted .rnbak; later menu JSON
Cloudflare Pages (later)	Static / PWA	QR view-only menu; owner dashboard web
SMS OTP (later)	Owner auth	Owner login - not Azure SQL

Local/CI only: SQLite file for Cloud API tests.

2. By product capability

Capability	Status	Services
Cloud backup / new-PC recover	Shipped (API)	App Service + Postgres (metadata) + R2 (ciphertext)
Owner dashboard	Planned	Same API + Postgres projections; Pages PWA; OTP
QR view-only menu	Planned	R2 + Pages (not Postgres on every guest scan)
Classic POS / Captain	Local only	No cloud services

3. Explicitly out

Do not use	Why
Azure SQL (Serverless or provisioned)	Cost; wrong fit for metadata/projections
Azure Blob / AWS S3 instead of R2	R2: no egress fee; already integrated
Go rewrite / AWS or Oracle now	Risk; little saving vs Postgres on Azure
Store .rnbak bytes in Postgres	Blobs stay on R2

4. Trust boundary

POS SQLite = source of truth for billing. Cloud stores ciphertext + metadata only. Recovery key held by owner; server stores verifier + wrapped key only.

5. Open questions

1) Postgres: public + firewall vs Private Endpoint + VNet? 2) Single-AZ B1ms OK for v1, or zone redundancy day one? 3) Key Vault mandatory for R2 + Postgres secrets? 4) Is R2 acceptable beside Azure Central India for ciphertext?

6. Ask

Approve: Linux App Service + PostgreSQL Flexible + R2 for backup now and owner dashboard / QR later. Reject: Azure SQL for production metadata; AWS/Oracle rewrite before paying Premium clients.

Detail plan: docs/RESTONITI_AZURE_POSTGRES_ARCHITECTURE_PLAN.pdf